

5CM506 Data Driven Systems

Tutorial and Exercise **T-SQL: Cursors**

Author: Dr. Maqbool Hussain
University of Derby

Displaying All Products of Type 'Customer Accounts' Using a Cursor

Objective

This tutorial will guide you through retrieving and displaying all products classified as '**Customer Accounts**' from a **Test** database.

Key Steps

- **Identify the relevant schema** that contains product information.
- **Determine the table** that holds 'Customer Accounts' product type.
- **Extract product type details** and locate associated product records.
- **Use a cursor** to iterate through product types and fetch product details if 'Customer Accounts' exists.

Steps to Implement

Step 1: Open Query Editor from SQL Server Managment Studio

- Launch your SQL query editor by clicking **New Query** window.

Step 2: Select the Database

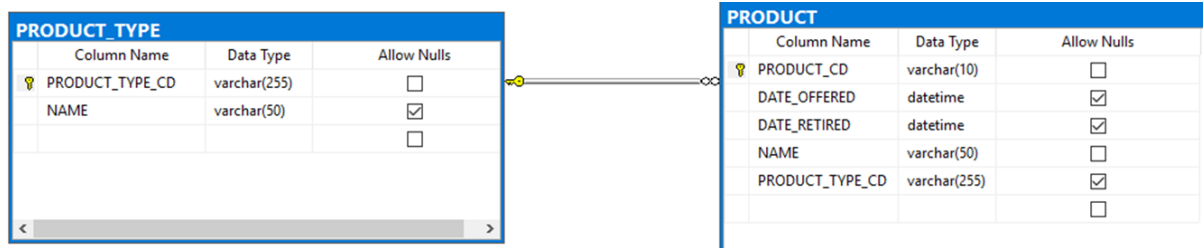
- Switch to the **Test** database, which contains the required product information.

USE Test;

Step 3: Inspect the Tables

- Verify the structure and contents of the relevant tables:
 - Check the schema
 - For data use queries
- **PRODUCT_TYPE** table contains product type details.
- **PRODUCT** table contains product details and links to **PRODUCT_TYPE** via **PRODUCT_TYPE_CD**.

Schema:



Queries for data:

```
SELECT * FROM PRODUCT_TYPE;
SELECT * FROM PRODUCT;
```

	PRODUCT_TYPE_CD	NAME
1	ACCOUNT	Customer Accounts
2	INSURANCE	Insurance Offerings
3	LOAN	Individual and Business Loans

	PRODUCT_CD	DATE_OFFERED	DATE_RETIRED	NAME	PRODUCT_TYPE_CD
1	AUT	2000-01-01 00:00:00.000	NULL	auto loan	LOAN
2	BUS	2000-01-01 00:00:00.000	NULL	business line of credit	LOAN
3	CD	2000-01-01 00:00:00.000	NULL	certificate of deposit	ACCOUNT
4	CHK	2000-01-01 00:00:00.000	NULL	checking account	ACCOUNT
5	MM	2000-01-01 00:00:00.000	NULL	money market account	ACCOUNT
6	MRT	2000-01-01 00:00:00.000	NULL	home mortgage	LOAN
7	SAV	2000-01-01 00:00:00.000	NULL	savings account	ACCOUNT
8	SBL	2000-01-01 00:00:00.000	NULL	small business loan	LOAN

Step 4: Declare Cursor for Product Types

- We define a cursor to iterate through all product types.

```
DECLARE prodType_cursor CURSOR FOR
    SELECT PT.PRODUCT_TYPE_CD, PT.NAME FROM PRODUCT_TYPE PT;
```

Step 5: Declare Variables to Hold Cursor Data

- We create variables to store product type details while iterating.
- Ensure the types conform to the type of columns in PRODUCT_TYPE_CD

```
DECLARE @prdType VARCHAR(255), @prdName VARCHAR(50);
```

Step 6: Open Cursor and Fetch First Row

- We initialize the cursor and fetch the first row.

```
OPEN prodType_cursor;  
  
FETCH NEXT FROM prodType_cursor  
    INTO @prdType, @prdName;
```

Step 7: Iterate Through Product Types

- We iterate through the cursor, checking for 'Customer Accounts'.

```
WHILE @@FETCH_STATUS = 0  
BEGIN  
    -- Check if the product type is 'Customer Accounts'  
    IF TRIM(@prdName) = 'Customer Accounts'  
    BEGIN  
        -- Retrieve and display all related products  
        SELECT * FROM PRODUCT WHERE PRODUCT_TYPE_CD = @prdType;  
        BREAK; -- Exit loop once found  
    END  
  
    -- Fetch next row if 'Customer Accounts' is not found  
    FETCH NEXT FROM prodType_cursor  
        INTO @prdType, @prdName;  
END
```

Step 8: Close and Deallocate Cursor

- After completing the iteration, we clean up the cursor.

```
CLOSE prodType_cursor;  
DEALLOCATE prodType_cursor;
```

Step 9: Execute the code and observe the results

Above code in One place:

```
USE Test;

DECLARE prodType_cursor CURSOR FOR
    SELECT PT.PRODUCT_TYPE_CD, PT.NAME FROM PRODUCT_TYPE PT;

DECLARE @prdType VARCHAR(255), @prdName VARCHAR(50);

OPEN prodType_cursor;

FETCH NEXT FROM prodType_cursor
    INTO @prdType, @prdName;

WHILE @@FETCH_STATUS = 0
BEGIN
    IF TRIM(@prdName) = 'Customer Accounts'
    BEGIN
        SELECT * FROM PRODUCT WHERE PRODUCT_TYPE_CD = @prdType;
        BREAK;
    END

    FETCH NEXT FROM prodType_cursor
        INTO @prdType, @prdName;
END

CLOSE prodType_cursor;
DEALLOCATE prodType_cursor;
```

Result:

	PRODUCT_CD	DATE_OFFERED	DATE_RETIRED	NAME	PRODUCT_TYPE_CD
1	CD	2000-01-01 00:00:00.000	NULL	certificate of deposit	ACCOUNT
2	CHK	2000-01-01 00:00:00.000	NULL	checking account	ACCOUNT
3	MM	2000-01-01 00:00:00.000	NULL	money market account	ACCOUNT
4	SAV	2000-01-01 00:00:00.000	NULL	savings account	ACCOUNT

Exercises

Exercise 1: Modify the Code to Meet the Requirements

Task

Modify the code from the tutorial to display all products of type '**Customer Accounts**', but only include the following columns in the output:

- **Product Type Name** (from `PRODUCT_TYPE`)
- **Product Name** (from `PRODUCT`)
- **Product Code** (from `PRODUCT`)

Instructions

- Use a cursor to iterate through product types.
- Identify and select products where the product type is '**Customer Accounts**'.
- Display only the specified columns in the result.
- Close and deallocate the cursor after execution.

Hint: Only query is required to be changed.

Expected Output Format

	Product Type Name	Product Name	PRODUCT_CD
1	Customer Accounts	certificate of deposit	CD
2	Customer Accounts	checking account	CHK
3	Customer Accounts	money market account	MM
4	Customer Accounts	savings account	SAV

Exercise 2: Modify the Code to Meet the Requirements

Task

Modify the existing code to display a **custom message** when a product meets the following conditions:

1. **Product Type Name** is 'Customer Accounts'
2. **Product Code** is 'CD'

Custom Message Format:

Product Type Name: Customer Accounts with **Product Code:** CD and
Product Name: *[corresponding product name]* is found.

Note: The value inside [] should be dynamically picked from an internal variable reflecting the product name.

Instructions

1. Use a **cursor for product types** to iterate through all product types.
2. If a product type is 'Customer Accounts', proceed to the next step.
3. Use a **nested cursor for products** to iterate through all products of that type.
4. If a product has **Product Code = 'CD'**, format and print the required message.
5. Close and deallocate the **inner product cursor** after execution.
6. Close and deallocate the **main product type cursor** after execution.

Hint

- Replace the **SELECT** statement with an **inner cursor** for iterating through products.
- Use the **PRINT** function to display the formatted message.

Expected Output Format

(Based on existing data, an example message could be:)

Product Type Name: Customer Accounts with Product Code: CD and Product Name: certificate of deposit is found.

Exercise 3 (Optional): Optimize the Solution Using a Single Cursor

Task

Refactor the solution from **Exercise 2** to achieve the same functionality **using only a single cursor** instead of nested cursors.

.